

Chosen ML Application - Social Media Recommendation

Part A - Identify the ML Problem

1. What is the application?

The application is a social media recommendation system. It is a type of system that is commonly used by social media platforms that takes data from the user and recommend specific posts, videos, marketing, and other content that is based on the user's interest

2. What problem are you trying to solve?

The problem we are trying to solve is that social media contents have broad topics and we want the user to feel that an app is consistent by making the content they are seeing most of the time captures their interests and needs.

3. What are the inputs?

The inputs are the user's liked posts, watched videos and how long they watched it, accounts they visit and follow. These activities serve as information of what the users are interested in and what recommended content would match them.

4. What is the expected output?

The expected output is to provide the users with a set of content that matches their interests. For example, when a user interacts with movie contents most of the time, the goal is to provide the user a set of posts, videos, and other content related to movies that are more likely to get an interaction from the user

5. What patterns or relationships might the ML system learn?

The system may learn that some users are connected to certain genres of content. For example, if a user regularly interacts with vehicle contents, the system may learn that the user is into vehicles. It can also detect which type of content the user skips and what videos the user watches more and what videos he skips.

6. What data would the system need?

The data that the system would be the user's search history, viewing history, liked posts and videos, contents they comment on, followed pages. It can also gather data from the content itself such as it types and where it is categorized. These examples of data will help the system recognize the user's pattern accurately.

Part B - Traditional Programming vs. Machine Learning

Traditional Programming

In Traditional Programming, the developer creates the rules or guidelines that the system needs to follow. For example, If a user likes many gaming videos, the program or the system could be instructed to show more gaming videos. The developer would have to manually create many rules for different types of users and content.

Input → Algorithm or Rules → Output

Machine Learning

In Machine Learning, the system is not given every rule manually. Rather, it receives data from past user actions and uses those examples to identify patterns. For instance, it can determine what kind of content a user is likely to interact with by looking at what they watch, like, and search for. After learning from the data, the system can use the pattern it found to suggest new content.

Input or Data → MLAlgorithm → Learned Pattern → Prediction

The main difference is that traditional programming relies on developer created rules. Likewise, machine learning makes predictions and finds patterns in data.

Part C - Create Your Own Mini Dataset

Time Spent Watching Videos and Recommendation Score

Time Spent Watching (minutes)	Recommendation Score
5	1
10	2
15	3
20	4
25	5

1. Identify the Patterns

The pattern is that the output increases by 1 whenever the input increases by 5.

2. Determine the missing output

If the input is 30 minutes, the output is 6.

3. Write the mathematical relationship/function.

The relationship is Output = Input / 5

Or $y = x / 5$

4. Explain how a machine learning algorithm could learn this relationship

The provided input and output examples can be compared by the algorithm to see how they change. It can infer from the examples that for every extra five minutes, the output rises by one. It can use the relationship to predict the output for a new input once it has learned this pattern.

Part D - Explain the Training Process

The machine learning algorithm is given data with examples of inputs and outputs during training. It examines these examples and searches for patterns among them. For instance, the software may discover that a user's interest in a particular kind of content may rise as they watch it more frequently. The algorithm makes predictions for new data using the patterns it has learned from sufficient examples.

Data → Training → Pattern Learning → Prediction

Short Reflection

I discovered that machine learning differs from traditional programming in that it doesn't always require a programr or developer to write a rule for every circumstance. Rather, it can identify patterns in the data and learn from examples. I can relate with this because every time I search on Facebook or TikTok, I notice that stores, merchandise, or suggestions that are relevant to my query start to show up in my feed. This demonstrates how the system can suggest items that I might find interesting based on my prior searches and activities.

Additionally, I discovered that machine learning can be helpful in simplifying and personalizing our tasks. The system can learn from each user's actions and make recommendations based on their interests rather than having to manually decide what content to display to each one. However, the system's recommendations are still dependent on the data it receives, so having sufficient and valuable data is crucial.